All code is in `Scripts/`, one script per question. Each script uses the general samplers in the project root (`SAMC.py`, built on `Metropolis_Hastings.py`) and saves its figure in `Figures/` and its numbers in `Results/`.

SAMC, as used in all four questions

All four questions use stochastic approximation Monte Carlo (SAMC; Liang, Liu and Carroll, 2007). Each question samples from a target $\psi(x) = \exp(-U(x)/\tau)$, where U is an energy and τ a temperature. The sample space is split into subregions $E_1, \ldots, E_m$ by a chosen quantity (tour length, energy, or $|t|$), and $J(x)$ denotes the subregion of x. Each subregion has a log weight θ_i, starting at 0. SAMC runs MH on the working density

$$p_\theta(x) \propto \exp\left(-U(x)/\tau - \theta_{J(x)}\right).$$

All proposals used here are symmetric, so a proposed state y is accepted with probability

$$\alpha = \min\left\{1, \exp\left[-\frac{U(y) - U(x)}{\tau} + \theta_{J(x)} - \theta_{J(y)}\right]\right\}.$$

After each step, with x_t the current state, every weight is updated as

$$\theta_i \leftarrow \theta_i + \gamma_t\left(\mathbf{1}\{i = J(x_t)\} - \pi_i\right), \qquad \gamma_t = \frac{t_0}{\max(t_0, t)},$$

where $\pi_i = 1/m$ is the desired share of time in each subregion. The weight of the subregion the chain is in goes up, which makes that subregion less attractive. A chain stuck in one mode or local minimum is therefore pushed out of it, and in the long run it spends about a $1/m$ share of time in every subregion it can reach.

The results are used in two ways.

Optimization (Q1, Q2). The lowest-energy state visited is saved.

Estimation (Q3, Q4). The samples follow the working density, not ψ. Each stored sample x_t gets the importance weight $w_t = \exp(\theta_{J(x_t)})$, using the weights at that iteration, and an expectation under ψ is estimated by the weighted average

$$\widehat{E}[h(x)] = \frac{\sum_t w_t\, h(x_t)}{\sum_t w_t}.$$

Q1. SAMC for 100-city tours

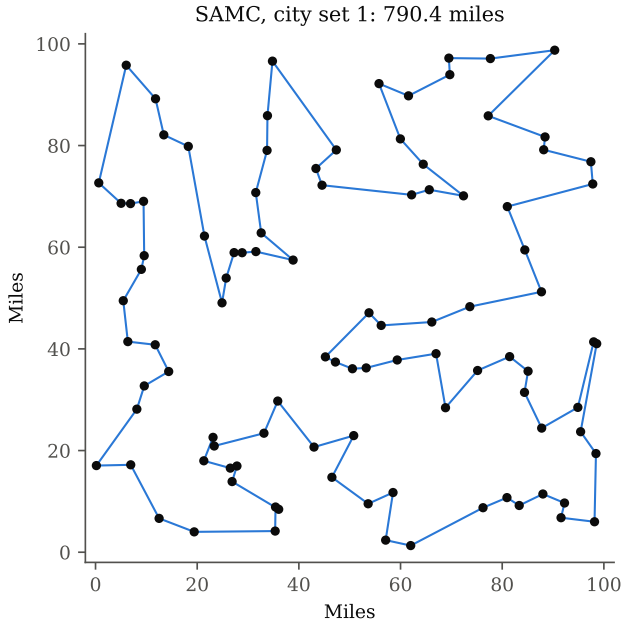
I interpreted the requested average as an average over independently generated city sets. I generated 20 sets, each containing 100 cities sampled uniformly from $[0, 100] \times [0, 100]$ miles, and ran SAMC once on each set from a random tour.

A tour is an ordering of all 100 cities. Its energy is its length L: the sum of Euclidean distances between consecutive cities, including the return from the last city to the first. Each proposal reverses one segment of the tour (a 2-opt move), so it keeps every city exactly once.

I split the tours by length into $m = 61$ subregions, using the levels $700, 705, \ldots, 995$ miles: $E_1 = \{L \leq 700\}$, 5-mile subregions in between, and $E_{61} = \{L > 995\}$. The shortest tour visited is saved.

Setting / result	**Value**
City sets / seed	20 / 546
Iterations per set	1,000,000
Temperature τ / gain constant t_0	2 / 5,000
Subregions	61, with 5-mile levels from 700 to 995 miles
Average shortest tour across 20 sets	**787.7 miles**

These are the shortest tours found within the stated budget. Their true optimality is not certified, so the average may be slightly above the average of the exact shortest tours.



Q2. Simulated stochastic approximation annealing for 100-city tours

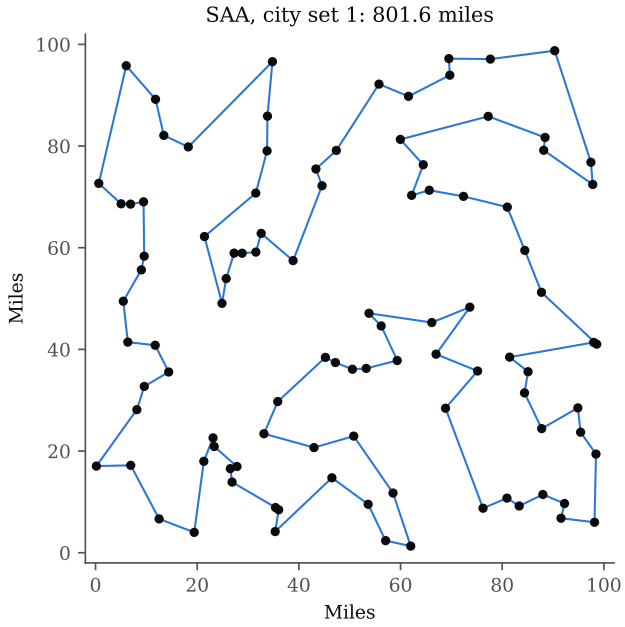
I used the same 20 city sets, the same 2-opt moves and the same 61 subregions as in Q1. Simulated stochastic approximation annealing (SAA; Liang, Cheng and Lin, 2014) is SAMC whose temperature is lowered during the run. At iteration t the working density is

$$p_{\theta,t}(x) \propto \exp\left(-L(x)/\tau_t - \theta_{J(x)}\right), \qquad \tau_t = T_{\text{high}}\sqrt{\frac{t_c}{\max(t_c, t)}},$$

so the temperature falls with the square root of the iteration number. The acceptance probability and the weight update are the SAMC ones, with τ_t in place of τ. The weights still push the chain out of subregions it stays in, while the falling temperature concentrates it on shorter tours. The shortest tour visited is saved.

Setting / result	**Value**
City sets / seed	20 / 546 (as in Q1)
Iterations per set	1,000,000
Cooling	$T_{\text{high}} = 10$, $t_c = 1{,}000$; τ falls from 10 to 0.32
Gain constant t_0 / subregions	5,000 / 61 (as in Q1)
Average shortest tour across 20 sets	**787.4 miles**

As in Q1, the shortest tours found are not certified optimal.



Q3. SAMC for the mean and covariance of the normal mixture

The target is $f(x) = \frac{1}{3}N(\mu_1, \Sigma_1) + \frac{1}{3}N(\mu_2, \Sigma_2) + \frac{1}{3}N(\mu_3, \Sigma_3)$ exactly as given. Its three modes are far apart, so a plain random-walk MH chain would rarely move between them.

I used the energy $U(x) = -\log f(x)$ with $\tau = 1$, so $\psi = f$, and split the plane by energy into $m = 14$ subregions with levels $2.5, 3.5, \ldots, 14.5$: $E_1 = \{U \le 2.5\}$, unit-width subregions in between, and $E_{14} = \{U > 14.5\}$. Each proposal is a normal random move. Because the chain spends about a 1/14 share of time in every subregion, including the high-energy ones between the modes, it crosses between the modes regularly.

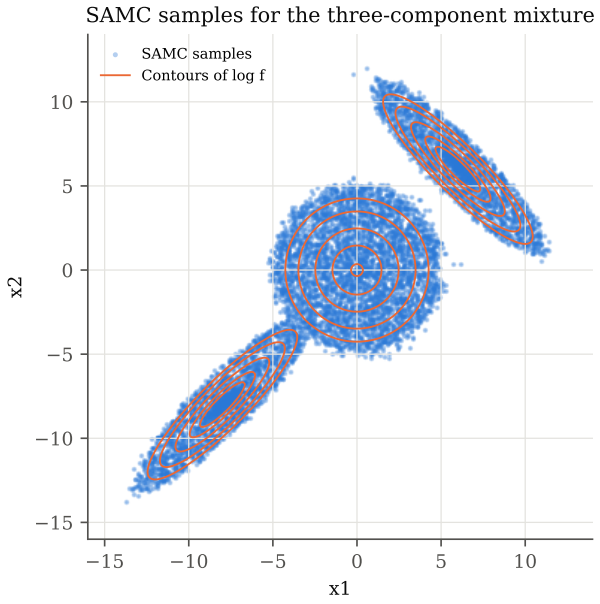
The mean and covariance are the weighted averages from the SAMC section:

$$\hat{\mu} = \frac{\sum_t w_t x_t}{\sum_t w_t}, \qquad \hat{\Sigma} = \frac{\sum_t w_t (x_t - \hat{\mu})(x_t - \hat{\mu})^T}{\sum_t w_t}.$$

The exact values are $\mu = \frac{1}{3}(\mu_1 + \mu_2 + \mu_3)$ and $\Sigma = \frac{1}{3}\sum_k (\Sigma_k + \mu_k \mu_k^T) - \mu\mu^T$, which give a direct check.

Setting	**Value**
Iterations / discarded / seed	4,000,000 / 400,000 / 546
Stored samples	Every 10th state after burn-in
Normal proposal standard deviation / t_0	1.5 / 1,000

Quantity	**SAMC estimate**	**Exact**
Mean μ	$(-0.649, -0.640)$	$(-0.667, -0.667)$
Covariance Σ	$\begin{pmatrix} 33.819 & 32.857 \\ 32.857 & 33.898 \end{pmatrix}$	$\begin{pmatrix} 33.889 & 32.889 \\ 32.889 & 33.889 \end{pmatrix}$



Q4. SAMC for p-values of the two-sample t-test

Data and p-value

The assignment does not supply data. I simulated $n_1 = 12$ observations from $N(0,1)$ and $n_2 = 12$ from $N(2,1)$, with seed 546; the values are saved with the results. The test statistic is the pooled two-sample t-statistic, and its observed value is $|t_{\text{obs}}| = 4.696$.

I evaluated the permutation p-value: the fraction of all ways of splitting the 24 values into groups of 12 and 12 whose $|t|$ is at least $|t_{\text{obs}}|$. Under the null hypothesis every split is equally likely. There are $\binom{24}{12} = 2{,}704{,}156$ splits, few enough to list them all, which gives the exact p-value as a check.

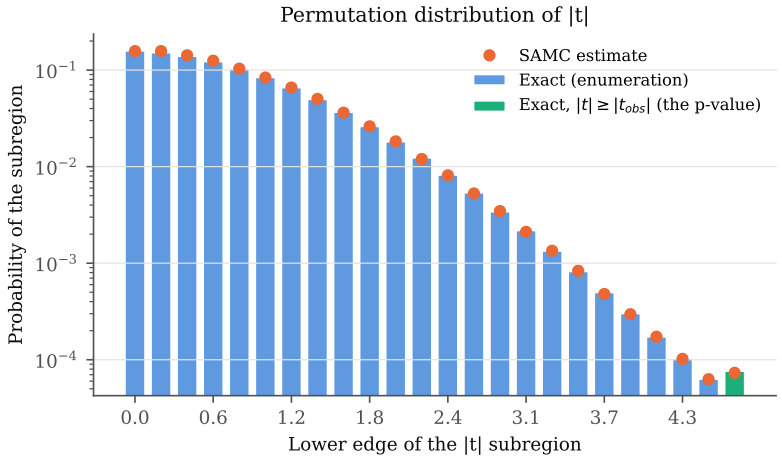
SAMC

Using SAMC to evaluate small p-values of resampling-based tests was proposed by Yu et al. (2011). The state is a split of the data. Each proposal swaps one randomly chosen member of group 1 with one randomly chosen member of group 2. The target is uniform on the splits, so the energy is 0 and the acceptance probability is $\min\{1, \exp(\theta_{J(x)} - \theta_{J(y)})\}$.

I divided the splits by $|t|$ into $m = 24$ subregions: 23 equal-width subregions covering $[0, |t_{\text{obs}}|)$, and the tail $E_{24} = \{|t| \geq |t_{\text{obs}}|\}$. Because the chain spends about a 1/24 share of time in every subregion, it visits the tail often even though the tail holds very few splits. The p-value is the weighted average from the SAMC section with $h(x) = \mathbf{1}\{x \in E_{24}\}$:

$$\hat{p} = \frac{\sum_t w_t \, \mathbf{1}\{x_t \in E_{24}\}}{\sum_t w_t}.$$

Setting / result	**Value**
Iterations / discarded / seed	1,000,000 / 100,000 / 546
Stored samples / gain constant t_0	Every 10th state after burn-in / 300
SAMC p-value	7.32×10^{-5}
Exact permutation p-value	7.69×10^{-5}



References

STAT 546, Fall 2026. Homework 4, Questions 1–4. The problems and requested algorithms are taken from the assignment.

F. Liang, C. Liu and R. J. Carroll (2007). Stochastic approximation in Monte Carlo computation. *Journal of the American Statistical Association*, 102, 305–320. DOI: 10.1198/016214506000001202.

F. Liang, Y. Cheng and G. Lin (2014). Simulated stochastic approximation annealing for global optimization with a square-root cooling schedule. *Journal of the American Statistical Association*, 109(506), 847–863. DOI: 10.1080/01621459.2013.872993.

K. Yu, F. Liang, J. Ciampa and N. Chatterjee (2011). Efficient p-value evaluation for resampling-based tests. *Biostatistics*, 12(3), 582–593. DOI: 10.1093/biostatistics/kxq078.